

PROGRAMMING FOR ARTIFICIAL INTELLIGENCE

Abdul Haseeb
BS(AI)-IV

Outlines

Programming Languages widely used for AI

Introduction to Python Programming
Language

Why Python Programming Language for AI

Introduction to Jupyter Notebook
(Notebook Environment)

Understanding Basic Python Syntax

Writing Basic Python Code (Print Function)

Programming Languages for AI

Python (Most
Widely Used)

R

Java

C++

Julia

Who created **Python** & when?



Guido van Rossum

1991

Invented and released
Python

What is **Python**?



Python is a high-level,
interpreted programming
language known for its

- ✓ Simple and readable syntax
- ✓ Versatility and broad appeal

What is **Python** used for?



Web
Development



Data
Science



Machine
Learning & AI



Why Python is Essential for AI



Easy to Learn & Use
Simple & Readable Syntax

Powerful Libraries

Great for Prototyping
Quick Development & Testing




TensorFlow


PyTorch


Scikit-Learn

Vast Community Support
Help from Developers Worldwide



Ideal for

Machine Learning



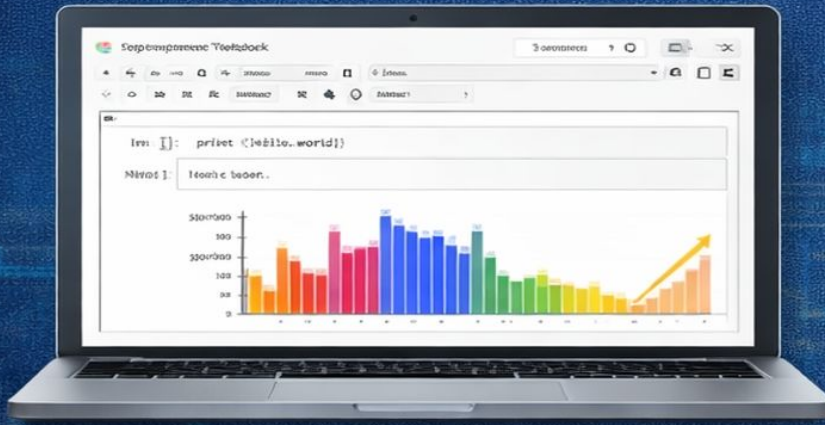
Deep Learning



Data Science



What is **Jupyter Notebook**?



An open-source web-based interactive computing environment used with **Python** and other languages

Advantages of **Jupyter Notebook**



Interactive Code & Visualizations



Combines Code, Text, & Output



Easy Experimentation



Popular for Data Science and AI

What is the **print()** function?

The `print()` function in Python is used to **display** output to the **console or terminal**.

Syntax:

```
print(*objects, sep=' ', end='\n', file=sys.stdout,
```

Basic Usage

You can pass multiple arguments **separated by commas**:

```
print("The sum of", 5, 'and', 3, 'is", 5+3)
```

Output: The sum of **5 and 3 is 8**

```
print("Python", "Data", "Science", sep='- ' - '')
```

Output: Python - Data - Science

Output: Python - Data - Science

Key arguments:

- **sep:** Separator between arguments
- **end:** Character appended at the end

Examples of **print()** Function

```
print("Hello, World!")
```

Hello, World!

```
print("Python is great!", "Let's learn!")
```

42 3.14 True

Output: The sum of **5 and 3 is 8**

Using **f-strings** for Formatted Output

Create formatted strings using **f-strings**:

```
name = "Alice"
```

```
age = 25
```

```
printf("My name is {name} and I am {age} years old.")
```

Output: My name is Alice and I am **25 years old**.

Key arguments:

- **sep:** Separator between arguments (default: ' ')
- **end:** Character appended at the end (default: '\n')